COHOMOLOGY WITH SUPPORT AND THE TRIANGLE RELATING IT TO THE COHOMOLOGY OF THE OPEN COMPLEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the resolution, the short exact sequence of complexes and the resulting triangle are all accounted for inside the counted span.

Editorial note. The label lemma-cohomology-with-support-triangle occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Sheaf with support in a closed subscheme and its derived version. Same chapter, the sheaf H_Z(F) of sections supported in Z and the derived functors R H_Z. Supplies the object i_* R H_Z(F) appearing in the distinguished triangle.
- Short exact sequence relating the sheaf, its sections with support and the restriction to the complement. Same chapter, the short exact sequence 0 -> i_*H_Z(F) -> F -> j_*(F|_U) -> 0 for a closed immersion i and the open complement j. Supplies the short exact sequence of complexes from which the triangle is produced.
- Injective resolutions and the distinguished triangle attached to a short exact sequence of complexes. Same chapter together with the chapter on Derived Categories, the passage from a short exact sequence of complexes to a distinguished triangle in the derived category. Supplies the step producing the distinguished triangle and hence the exact sequence and the isomorphisms of the statement.

Source reference. Original section 014Z: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14101–14101. Cross-chapter navigation pointer invoked at proof line 14101, where the author refers to the construction attaching a distinguished triangle to a short exact sequence of complexes of sheaves. The target chapter is not staged in this card's source tree, so the pointer records the invoking

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line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of schemes. Let $j : U \to X$ be the inclusion of the complement of Z. Let $\mathcal{F}$ be an abelian sheaf on $X_{étale}$. There is a distinguished triangle*

$$i_*R\mathcal{H}_Z(\mathcal{F}) \to \mathcal{F} \to Rj_*(\mathcal{F}|_U) \to i_*R\mathcal{H}_Z(\mathcal{F})[1]$$

in $D(X_{étale})$. This produces an exact sequence

$$0 \to i_*\mathcal{H}_Z(\mathcal{F}) \to \mathcal{F} \to j_*(\mathcal{F}|_U) \to i_*\mathcal{H}^1_Z(\mathcal{F}) \to 0$$

and isomorphisms $R^pj_*(\mathcal{F}|_U) \cong i_*\mathcal{H}^{p+1}_Z(\mathcal{F})$ for $p \geq 1$.

Proof. To get the distinguished triangle, choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$. Then we obtain a short exact sequence of complexes

$$0 \to i_*\mathcal{H}_Z(\mathcal{I}^\bullet) \to \mathcal{I}^\bullet \to j_*(\mathcal{I}^\bullet|_U) \to 0$$

by the discussion above. Thus the distinguished triangle by Derived Categories, Section 014Z. □